

A comprehensive revision has been made in this revised manuscript, as follows:

1. The authors have reviewed the entire manuscript and corrected minor grammatical and spelling errors.
2. The limitations and future work sections have been carefully revised based on the reviewers' comments.
3. The consistency of terminology, model structure, and phrasing throughout the manuscript has also been carefully reviewed.

Moreover, all supporting resources, including the source code, simulation files for impedance/admittance scanning, and datasets, have been made publicly available to support reproducibility.